

GAP-M: Governance-Aligned Post-deployment Monitoring

A machine-readable crosswalk of NIST AI RMF, NIST AI 800-4, ONC HTI-1 predictive-DSI source attributes, and CISA CPG 2.0

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Abstract

This report describes GAP-M, an open, machine-readable set of 32 suggested post-deployment monitoring practices for predictive AI systems. It also describes a crosswalk that links each practice to elements of four U.S. federal sources of different kinds: voluntary guidance (the 72 subcategories of the NIST AI Risk Management Framework); a report on monitoring challenges (NIST AI 800-4: 6 monitoring categories, 19 category-specific and 13 cross-cutting gaps and barriers); a regulatory disclosure requirement for certified health IT (the 31 predictive decision support intervention source attributes of 45 CFR 170.315(b)(11)(iv)(B)); and voluntary cybersecurity practices (the 34 goals of CISA CPG 2.0). The 283 links use NIST IR 8477's supportive-relationship vocabulary, and every one of the 175 mappable source elements has a recorded disposition. At least one practice is linked to 149 elements (85.1%). A necessity audit tested every link that claimed a practice was necessary against the exact source text. After a second, stricter pass, 14 of 107 claims are supported, 84 were downgraded, and 9 remain unresolved and are kept as conditional example_of mappings. A tooling-assisted relevance review of the 210 original example_of links removed 31 and reworded 9, leaving 283 links. No necessity claim to an AI 800-4 element survived, because that report describes challenges and sets no requirements. Mapping coverage is a property of the crosswalk, not of any organization; it does not demonstrate regulatory compliance or monitoring effectiveness.

1. Why a crosswalk

Post-deployment monitoring is addressed, in different ways, by several federal documents. The NIST AI RMF describes the outcome that post-deployment monitoring plans are implemented (MANAGE 4.1). NIST AI 800-4 reports that practitioners lack trusted monitoring standards and find baselines, ground truth, and longitudinal tracking hard to sustain. ONC's HTI-1 rule requires certified health IT to support source attributes for predictive decision support interventions, including validity and fairness in local data. CISA's CPG 2.0 sets out voluntary baseline cybersecurity practices for critical infrastructure, including health care. In the texts used here, none of the four sources cites another. These texts are the AI RMF 1.0 document, the AI 800-4 report, the eCFR text of 170.315(b)(11), and the CPG 2.0 goal page. The AI RMF refers to the NIST Cybersecurity Framework, which is not one of the four, and AI 800-4 cites a different NIST AI report (AI 100-2). The HTI-1 preamble was not reviewed for this work and may discuss NIST guidance. No element-level correspondence among the four was found in the texts reviewed. GAP-M offers one set of suggested practices, plus a typed, machine-readable record of how each practice relates to each source element.

2. Four kinds of source

Source	Kind of statement	Who it obligates	Elements used
NIST AI RMF 1.0 (NIST AI 100-1)	Voluntary guidance: risk-management outcomes	No one (voluntary)	72 subcategories
NIST AI 800-4	Report on monitoring challenges: categories, gaps, barriers	No one (describes challenges; sets no requirements)	6 categories; 19 category and 13 cross-cutting gaps/barriers
45 CFR 170.315(b)(11)(iv)(B) (ONC HTI-1)	Regulatory disclosure requirement for certified health IT	The certified Health IT Module and its developer, not deploying organizations	31 source attributes
CISA CPG 2.0	Voluntary practices with recommended actions	No one (voluntary)	34 goals
GAP-M controls (this work)	Suggested implementation practices	No one; not endorsed by any agency	32 controls

The crosswalk interprets a link differently for each kind of source. An AI RMF subcategory is an outcome an organization may choose to pursue. An AI 800-4 gap or barrier is a challenge practitioners report; a practice can respond to it but cannot satisfy it. An HTI-1 source attribute is information a certified Health IT Module must support. The obligation sits with the module and its developer, and GAP-M links only describe how a deploying organization's monitoring can supply or check attribute content. A CPG 2.0 goal is a voluntary practice with a stated scope. GAP-M's own controls are suggested implementation practices, not requirements.

All four sources are U.S. government works. AI RMF text is parsed from the PDF layout, and line-break hyphens are resolved by a documented rule. AI 800-4 table text is transcribed and checked verbatim against the PDF (55 strings). HTI-1 text comes from the eCFR point-in-time XML dated 2026-10-01; the section was last amended 2025-10-01. The eCFR is an editorial compilation, not an official legal edition of the CFR. CPG 2.0 goals are parsed from an archived copy of the CISA page whose parsed records hash identically to those parsed from the live page on 2026-10-05; the release includes the parsed records, not the archived page. NIST AI 800-4 states that it "is focused on generative AI systems, however the concepts are relevant to other types of AI and machine learning" (p. 4, note 4); GAP-M applies its

categories and challenges to predictive systems on that basis.

3. Method

3.1 Suggested practices

GAP-M's 32 controls are original. They are grouped into 8 families: one governance family, and families that follow AI 800-4's six monitoring categories, with fairness as its own family under Functionality because AI 800-4 has no fairness category (GOV (7); FUN (6); FAIR (2); OPS (4); HF (3); SEC (4); CMP (3); LSI (3)). Each control states an objective, activity, evidence artifact, cadence, owner role, and a one-sentence core activity. Only the core counts when necessity is judged.

3.2 Mapping vocabulary and the necessity audit

Every link reads control supports element, with one NIST IR 8477 property. **integral_to**: the element cannot be achieved without the control's core activity. **precedes**: the core must be achieved first. **example_of**: the core is one way to achieve the element. Every link that claimed **integral_to** or **precedes** was audited against the exact source text, with a rule for each kind of source. For AI RMF, necessity is supported only where the subcategory's own words name the core activity. For AI 800-4, necessity is never supported, because a challenge has no completion condition. For HTI-1, necessity is supported only where no party other than the deploying organization can produce the attribute's value, and then only for a populated value, not for certification. For CPG 2.0, necessity is supported only where the goal's recommended action, within its stated scope, necessarily reaches the AI system. Links whose necessity depends on a fact or judgment only the author or a deploying organization can supply are marked **unresolved** and held at **example_of**. The pipeline now refuses any **integral_to** or **precedes** link without a supported audit entry, and any **example_of** rationale that asserts necessity.

3.3 Relevance review of **example_of** links

The 210 links that were **example_of** from the start were reviewed by the AI assistant against the exact source text and the control's own text. Each was checked for relevance (does the control's activity bear on the element as written), source interpretation (is the element's subject, phase, and scope read correctly), and unsupported claims (does the rationale say anything the control does not). A link failing relevance, or failing the other two in a way rewording could not fix, was removed. Removals were not offset to preserve coverage: an element that lost its only link received a disposition instead. This review is tooling-assisted. It is not independent expert validation and not the author's review.

3.4 Dispositions and GAP-M change classes

Every mappable element is either linked or given a disposition with a reason: **uncovered** (relevant to post-deployment monitoring, but no GAP-M control substantively supports it), **prerequisite** (a hosting-environment baseline GAP-M assumes), or **out of scope** (a design-time, cultural, or research-governance outcome that monitoring evidence does not show). GAP-M also sorts the HTI-1 attributes into four analytic change classes (Section 5). These classes are GAP-M's, not the regulation's.

4. Results

What coverage means. A mapped element has at least one GAP-M control linked to it. Mapping coverage is a property of the crosswalk, not of any organization: it does not show that an organization complies with a regulation, meets a framework outcome, or monitors effectively.

Table 1. Mapping coverage of each source after both reviews. Coverage fell during review because weak links were removed, not replaced.

Source	Elements	Mapped	With integral_to	Not mapped	Links
AI RMF 1.0 subcategories	72	61 (84.7%)	7 (9.7%)	2 uncovered, 9 out of scope	114
AI 800-4 categories + challenges	38	36 (94.7%)	0 (0.0%)	1 uncovered, 1 out of scope	58
HTI-1 source attributes	31	31 (100.0%)	1 (3.2%)	0	88
CPG 2.0 goals	34	21 (61.8%)	5 (14.7%)	13 prerequisite	23
Total	175	149 (85.1%)	13 (7.4%)	3 uncovered	283

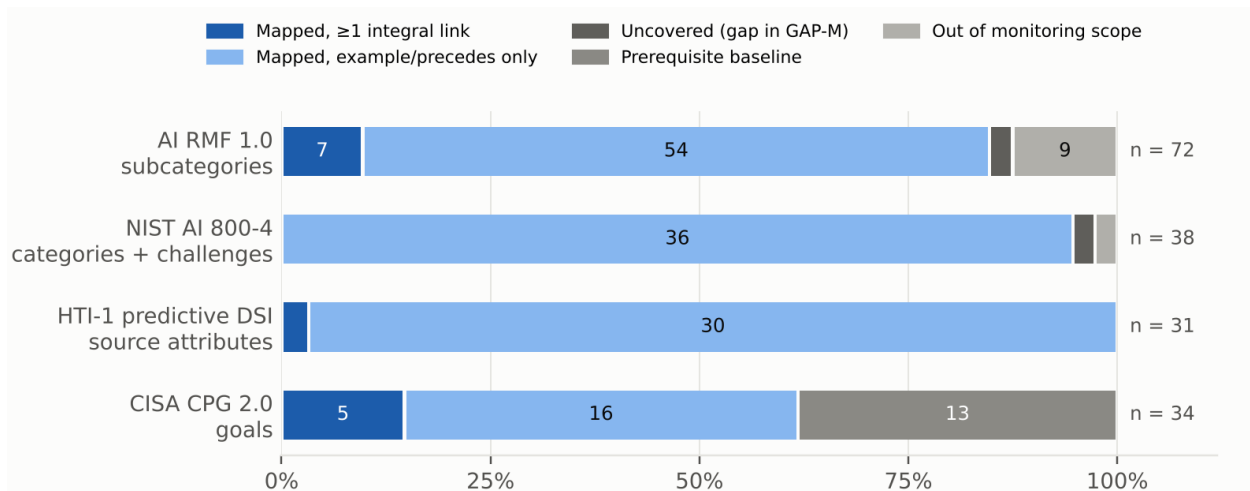
Table 2. Necessity audit of the 107 links that claimed integral_to (100) or precedes (7).

Source kind	Supported	Downgraded	Unresolved
Guidance (AI RMF)	8	21	3
Monitoring challenges (AI 800-4)	0	31	0
Disclosure requirement (HTI-1)	1	26	4
Voluntary practice (CPG 2.0)	5	6	2
Total	14	84	9

Table 3. Link ledger: every link of the original build by its original property. Current total 283 = 317 – 34.

Original property	Original	Kept as is	To example_of	Conditional (unresolved)	Removed (AI review)	Removed (author)
integral_to	100	14	77	8	1	0
precedes	7	0	5	1	0	1
example_of	210	178 (8 reworded)	—	—	31	1
Total	317				32	2

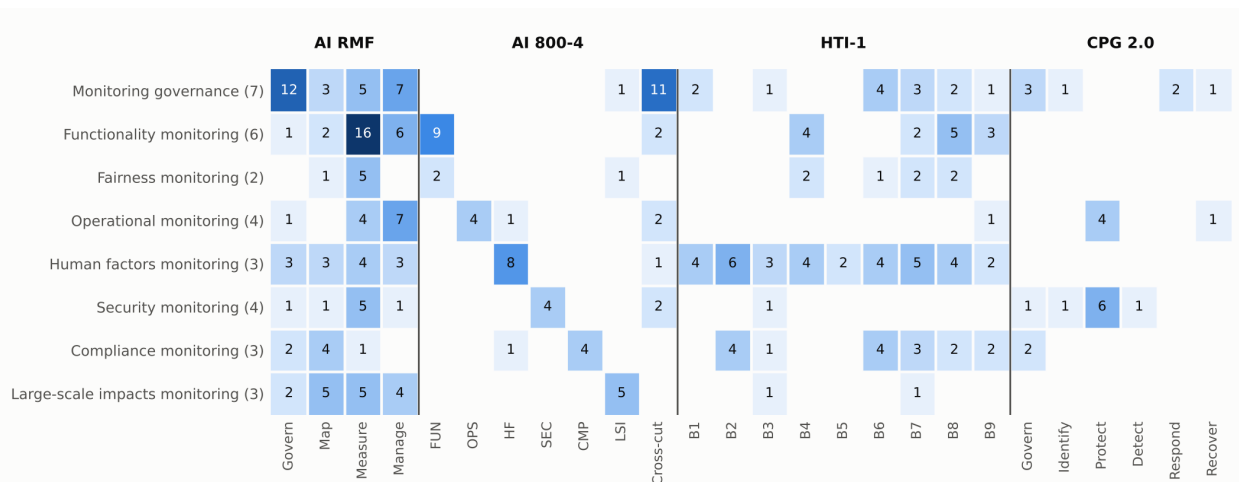
After both reviews the crosswalk holds 14 integral_to, 0 precedes, and 269 example_of links. Most links therefore say a practice is one way to pursue an element, not that it is required. The 14 remaining necessity claims are those where the source names the control's core activity, or where a CPG goal's scope necessarily reaches the AI system: AI RMF: MANAGE 4.1 (GM-GOV-01), GOVERN 1.6 (GM-GOV-02), MEASURE 4.3 (GM-FUN-05), MANAGE 2.4 (GM-FUN-06), MEASURE 3.3 (GM-HF-01), GOVERN 1.1 (GM-CMP-01), MANAGE 4.3 (GM-LSI-01), MEASURE 3.3 (GM-LSI-03); CPG 2.0: 1.D (GM-GOV-03), 3.Q (GM-OPS-01), 3.N (GM-OPS-03), 2.B (GM-SEC-02), 3.H (GM-SEC-03); HTI-1: (B)(8)(ii) (GM-FUN-02). A second pass rechecked every remaining necessity claim against a narrowed control core and changed 7 of them: 5 downgraded where a plausible alternative exists, and 2 held as unresolved where necessity turns on how the source is read (docs/MAPPING_REVIEW.md, section 11). The 9 unresolved links are kept as conditional example_of mappings. Their necessity is unresolved; the author reviewed each and accepted it as a conditional mapping. The condition under which each would be necessary is listed in docs/MAPPING_REVIEW.md. Controls carry a median of 7 links (range 4–38). 27 of 32 controls link to at least three of the four sources.



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Figure 1. Share of each source's elements by disposition and strongest link property, after both reviews.

Uncovered, a known gap in GAP-M v0.1.0 (3): MAP 3.2, MEASURE 2.12, XC IOC B1. **Out of monitoring scope** (10): GOVERN 2.3, GOVERN 3.1, GOVERN 3.2, GOVERN 4.1, MAP 1.2, MAP 1.3, MAP 4.1, MEASURE 2.2, MANAGE 3.2, LSI B1. **Prerequisite baseline** for the systems hosting the AI (13): CPG 2.C, 2.D, 2.E, 3.A, 3.B, 3.C, 3.I, 3.J, 3.L, 3.M, 3.R, 3.S, 4.A. Each carries its reason in data/crosswalk/gapm_dispositions.csv.



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Figure 2. Links by GAP-M control family (rows; control count in parentheses) and source group (columns).

5. HTI-1 source attributes and GAP-M change classes

The regulation's only language about currency is 170.315(b)(11)(v)(A)(1). It requires the Health IT Module to give access to "complete and up-to-date" descriptions of every attribute of a developer-supplied intervention. That requirement applies to all attributes alike, and the duty sits with the module and its developer. Paragraph (v)(A)(2) lets 11 attributes be shown as not available, and (v)(B) lets identified users record and change attributes. GAP-M's four change classes are an analytic classification of what most often changes an attribute's value after deployment. The source text does not establish them, and they do not decide which practices are necessary. The 15 attributes outside the set-at-release class are what earlier drafts called dynamic; that term is retired.

Table 4. HTI-1 attributes by GAP-M change class. Rows sum to the regulation's attribute count.

GAP-M change class	Attributes	With integral_to	With precedes	Example_of only	Not-available allowed
set at release	16	0	0	16	0
process description	4	0	0	4	2
new evidence possible	9	0	0	9	7
measured locally	2	1	0	1	2
Total	31	1	0	30	11

After both reviews, only B8.ii (validity of the intervention in local data) has a practice judged necessary: measuring validity in the deploying organization's own data. That necessity is for a populated value, not for certification. For (B)(8)(iv), local fairness, necessity is conditional, because the regulation does not define fairness and individual-fairness measures are an alternative to group comparisons. Outcome labels are not treated as a prerequisite for local fairness, because output-rate parity uses no labels. Whether they are a prerequisite for local validity is unresolved: label-free accuracy estimation exists [8], but the same work shows that target accuracy is identifiable only under assumptions on the shift, such as an unchanged outcome relationship $p(y|x)$, which local validation is meant to test. Whether the deploying organization must also record local values itself, rather than send them to the developer, is unresolved. Missing information is handled by attribute. The 11 attributes listed in (v)(A)(2) may be shown as not available. For the other 20, (v)(A)(1) expects complete and up-to-date descriptions, so GAP-M (GM-HF-03) records a missing value as a gap to raise with the developer, not as not available. Figure 3 shows every attribute-practice link.

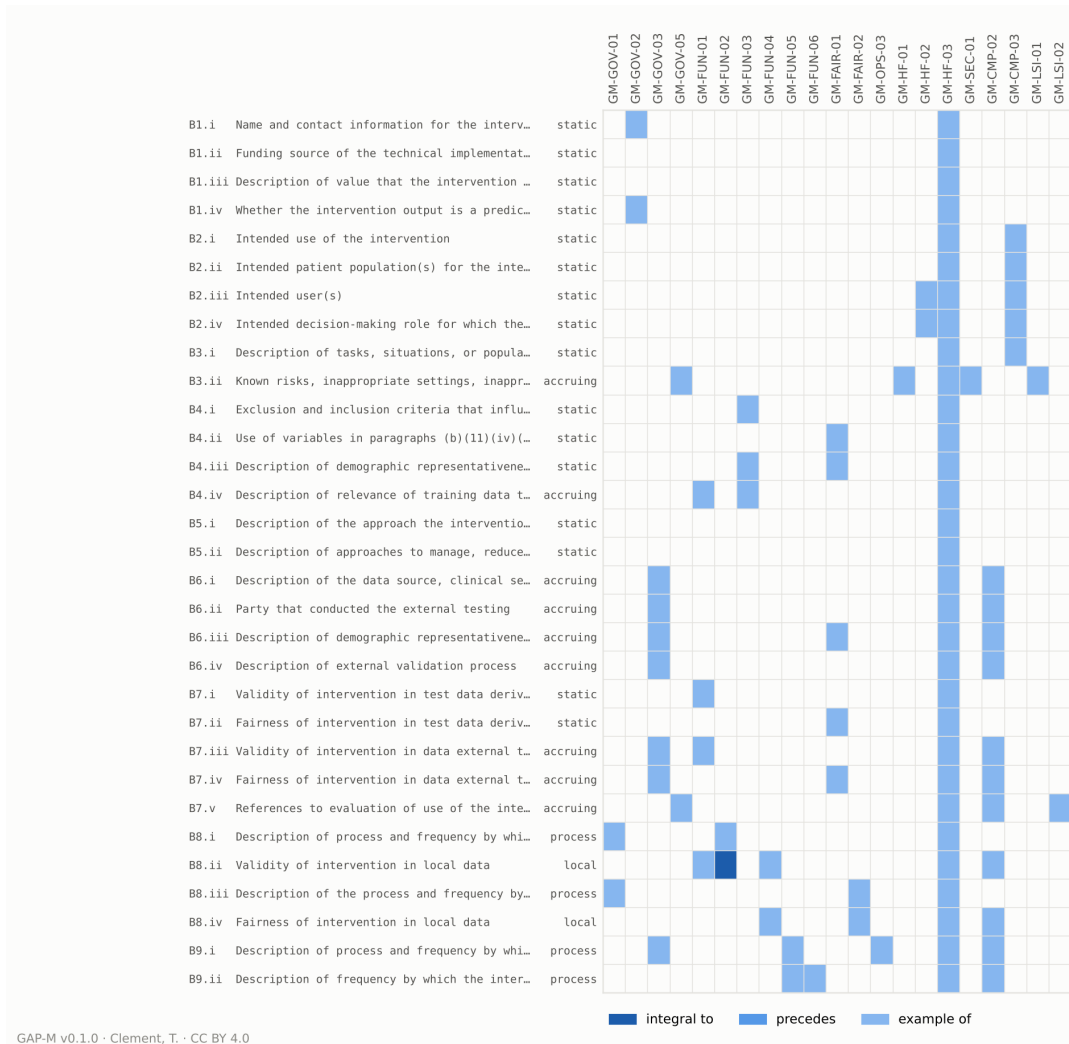


Figure 3. The 31 HTI-1 predictive-DSI source attributes (rows, with GAP-M change class) and the GAP-M controls linked to each, by IR 8477 property.

6. The suggested practices

Table 5. GAP-M controls and their link counts per source. Full definitions are in data/crosswalk/gapm_controls.csv.

ID	Control	RMF	800-4	HTI	CPG
GM-GOV-01	Monitoring plan and accountable owner	7	1	2	
GM-GOV-02	AI system and model inventory	2	1	2	1
GM-GOV-03	Developer monitoring obligations in contracts	4	1	7	2
GM-GOV-04	AI incident response integration	4	1		4
GM-GOV-05	External sharing of monitoring findings	3	3	2	
GM-GOV-06	Monitoring resources and competence	3	4		
GM-GOV-07	Independent assessment of monitoring	4	1		
GM-FUN-01	Performance baseline and deviation thresholds	6	2	4	
GM-FUN-02	Local validity monitoring	5	2	2	
GM-FUN-03	Input data and population drift detection	3	2	3	
GM-FUN-04	Outcome label acquisition	3	2	2	
GM-FUN-05	Longitudinal record and version comparison	3	2	2	

ID	Control	RMF	800-4	HTI	CPG
GM-FUN-06	Correction and deactivation triggers	5	1	1	
GM-FAIR-01	Subgroup performance baseline	3	1	5	
GM-FAIR-02	Local fairness monitoring	3	2	2	
GM-OPS-01	Central, tamper-protected monitoring logs	3	3		1
GM-OPS-02	Service level and cost telemetry	3	3		
GM-OPS-03	Change and configuration control for AI components	1	1	1	2
GM-OPS-04	Fallback and recovery	5			2
GM-HF-01	User feedback capture and adjudication	5	2	1	
GM-HF-02	Interaction and override telemetry	3	5	2	
GM-HF-03	Source-attribute currency and user transparency	5	2	31	
GM-SEC-01	Adversarial input, misuse, and anomalous behavior detection	2	2	1	1
GM-SEC-02	Vulnerability management for AI components	3	2		2
GM-SEC-03	Access control over models, configuration, and monitoring data	1	1		5
GM-SEC-04	Privacy-preserving handling of monitoring data	2	1		1
GM-CMP-01	Requirements register for the deployed system	2	2		2
GM-CMP-02	Source-attribute completeness audit	1	1	11	
GM-CMP-03	Out-of-scope and policy-violating use monitoring	4	2	5	
GM-LSI-01	AI incident and near-miss log with impact fields	4	2	1	
GM-LSI-02	Outcome and benefit measurement	6	2	1	
GM-LSI-03	Affected-community engagement and appeals	6	1		

7. Limitations

Mapping coverage does not demonstrate compliance with any regulation, achievement of any framework outcome, or effective monitoring. The links, the necessity audit, and the relevance review are judgment prepared with AI assistance; none is independent expert validation. 3 relevant elements are uncovered by any GAP-M control. The author has individually reviewed 98 of the 283 links; the other 185 have not been individually reviewed, and no inter-rater agreement has been measured. The practices have not been piloted. HTI-1 obligations fall on certified health IT, not on deploying organizations, and only paragraph (b)(11)(iv)(B) is mapped. AI 800-4 describes challenges, not requirements, and focuses on generative AI; GAP-M applies it to predictive systems. GAP-M's attribute change classes are analytic. The AI RMF Playbook and Generative AI Profile are not mapped. GAP-M has no official status. The full list is in docs/LIMITATIONS.md.

8. Verification status

Author's statement. Author review completed before release.

GAP-M v0.1.0 was prepared with substantial assistance from an AI assistant, Claude (Anthropic). I set the project's scope and target. The assistant wrote the code that retrieves, parses, and validates the sources and builds the documents; drafted the 32 practice definitions, every link and its rationale, the source-element dispositions, and the text of the README, technical report, and supporting documents; and carried out, with tooling support, a necessity audit of every link that claimed a practice was necessary and a relevance review of the 210 links that were example_of from the start.

I then reviewed the work in a guided process. For each item the assistant showed the exact source text, the practice's core, the proposed relationship and rationale, and a recommendation, and I recorded my decision. I reviewed all 32 practice definitions, accepting 18 as written and 14 with edits. I individually reviewed 100 links: all 14 integral_to links, all 9 conditional links, a random sample of 66 of the 262 other example_of links (25%, drawn once with seed 20261005), and 11 links in targeted checks. Of these I accepted 87 as written, accepted 11 with reworded rationales, and removed 2. I also ruled on every change made by the AI-assisted review: I confirmed its 32 removals and accepted 8 of its 9 reworded rationales (the ninth applied to a link I had removed). I made the open framing decisions recorded in docs/AUTHOR_DECISIONS.md, and I spot-checked named passages and counts in all four sources against the original documents. In nearly every case I accepted the assistant's recommended option, and I ruled on many links as groups after reading each one.

Separately from my review, the build pipeline runs automated checks that the assistant wrote and ran. They compare source element counts with the documents; check transcribed text verbatim, including 55 strings from NIST AI 800-4; compare the CPG 2.0 records against a hash of the live CISA page; enforce the crosswalk's consistency rules; validate every output against its JSON Schema and the OSCAL catalog against NIST's OSCAL 1.1.3 schema; and confirm that corrupted inputs are rejected (5 of 5 tests). A clean rebuild in the same environment reproduces every file byte for byte; a rebuild on a second computer reproduces the data and text exactly. These checks show that the package is internally consistent and faithful to the source text as parsed. They do not show that any mapping is correct, and they are not part of my review.

My review covered a defined subset. Of the 283 links in this release, 185 were not individually reviewed by me; they rest on the assistant's drafting and the AI-assisted reviews, and they are marked review_status: proposed and author_response: not_reviewed in the data. Sixteen of them received wording-only consistency edits that I approved without reviewing each link. Links that the necessity audit downgraded to example_of had a lighter relevance check than the rest: I reviewed 27 of the 82, through my random sample and one targeted check. My source check was a spot-check of named items; full verification of the transcribed text was done by the automated checks.

Neither my review nor the AI-assisted reviews is independent expert validation, and no inter-rater agreement has been measured. The practices have not been piloted in any deploying organization. GAP-M is independent work; it is not a product of NIST, ONC/ASTP, or CISA, and none of these agencies has reviewed or endorsed it. Mapping coverage does not demonstrate compliance with any regulation, achievement of any framework outcome, or effective monitoring. I take responsibility for the decisions recorded in docs/AUTHOR_REVIEW.md and docs/AUTHOR_DECISIONS.md.

9. Availability

Released under CC BY 4.0 (data, schemas, figures, report) and MIT (code) at <https://github.com/TosinClement/gap-m>, archived at Zenodo, DOI 10.5281/zenodo.23178136.

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